The Incumbency Paradox in Indian Elections

An Analysis of Incumbency effects on elections

Final Project Report - Group 043

Meta Information

Open Project Title	The Incumbency Paradox in Indian Elections – An Analysis of Incumbency effects on elections.
Group ID	043
Mentor Name	Mayank Jain
Group Members	Praveen (25b1037), Arjun (25b0963), Ruthvik (25b0981), Ram (25b1079).
Honor Code and Discussions	The submitted work has been reviewed by all group members. Apart from consultations with the mentor, no external discussions took place.
LLM Usage Links	**Chat links:** Chat 1, Chat 2, Chat 3, Chat4. **Automated agent report:** Not Used.
Video Presentation	Video Presentation Link
GitHub Repository	The Incumbency Paradox in Indian Elections
GitHub Handles and Initial Commits	Ruthvik – @ruthvikiit25iit (18 Oct 2026) Arjun – @arjun-v1157 (18 Oct 2026) Praveen – @praveenkumarkola06-dotcom (01 Nov 2026) Ram – @JaswanthRam007 (18 Oct 2026)
Contribution Breakdown	Unequal contribution (sample): 25b0963 – 100%, 25b0981 – 100%, 25b1037 – 100%, 25b1079 – 100%. LLM: 25%(collectively across all group members)

Technical Content

Core problem statement

This project investigates whether incumbent candidates have an electoral advantage over non-incumbent candidates in Indian Lok Sabha general elections. More specifically, to what extent is incumbency associated with a candidate's likelihood of winning an election and the margin of victory, after accounting for the influence of factors such as previous electoral performance and political party candidate selection? At first glance, comparing the win rates of incumbents and challengers might seem like a straightforward way to answer this question. However, the results can be misleading because political parties often choose to re-field candidates they believe have a good chance of winning.

Data source(s), Provenance, Scale, and Structure

The primary dataset is the **TCPD Lok Dhaba Election Dataset (TCPD-IED)**. The source file used in the project contains 91,669 candidate-election records and 44 columns, with observed years from **1962 to 2021**. The working analysis used poll_type = 0 for general-election records and excluded by-elections. Candidate-election records include information used to identify candidates, election years, incumbency status, electoral outcomes, and margins. Dataset portal: **click here**. This has been compiled by the Trivedi Centre for Political Data (TCPD) at Ashoka University from Election Commission of India reports, with incumbency and political-career variables per candidate.

Tools and software packages

The analysis workflow uses **Python 3**, with **Pandas** for reading, filtering, grouping, and tabulating candidate-election records, and **Matplotlib** for graphs. Used Numpy, SciPy , and stats to calcualte Confidence Intervals and to implement regression analysis on the intended variables. The complete directory containing the datasets and analysis scripts is maintained in a Git repository, enabling version control and retrieval of previous versions in case of errors or unintended modifications.LaTeX is used for report preparation.

India Focus

The dataset is India-specific and is derived from Election Commission of India (ECI) reports, compiled and curated by the Trivedi Centre for Political Data (TCPD). It contains records of candidates who contested Lok Sabha elections across India, covering the period from 1962 to 2021.

Challenges and Results Achieved

The main challenges involved filtering general elections correctly, classifying incumbents, verifying winner labels and denominators, tracking candidates across elections, and handling missing or inconsistent electoral margins and candidate matches.

Win-rate analysis. Incumbents recorded a win rate of 57.90% (2,837/4,900), compared to 40.23% (5,171/12,855) for non-incumbents. The difference was 17.67 percentage points (95% CI: 16.04–19.29). The respective 95% confidence intervals were 56.51–59.27% and 39.38–41.08%. The year-wise gap varied across elections rather than increasing uniformly.

Victory margins. Incumbent winners had a mean margin of 16.75% and median of 13.19%, compared to 15.29% and 11.81% for non-incumbent winners. The mean difference was 1.47 percentage points.

Regression analysis. Simple linear regression on 6,168 same-constituency observations gave $\hat{Y} = 0.2690 + 0.1916X$ ($R^2 = 0.0389$), indicating a positive but weak association between previous and subsequent margins. Multiple linear regression on 5,051 observations gave $\hat{Y} = -4.4949 - 2.5948I + 0.2182M + 0.1829P$ ($R^2 = 0.0816$). The adjusted incumbency coefficient was -2.5948 (95% CI: -4.1829 to -1.0067; $p = 0.001368$), indicating a negative association after accounting for previous margin and party vote share.

Overall, incumbents show higher observed win rates and larger winning margins, while the adjusted regression reveals a different relationship. These observational results do not establish causation. The inspected dataset covers elections up to 2021.

FAQ

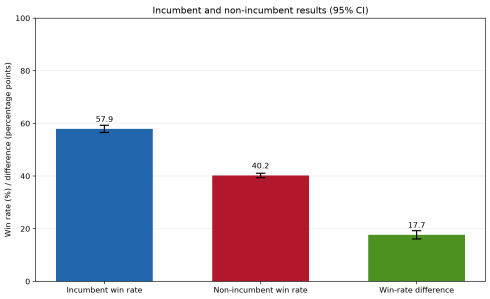
Do incumbent candidates really have an electoral advantage?

The first step is to compare the observed winning rates of candidates classified as incumbents and non-incumbents. The analysis uses candidate–election records from Lok Sabha general elections, retaining `poll_type = 0`; by-elections are excluded because they fill individual vacancies and may occur under different electoral circumstances. A candidate's record is counted as a win when `position = 1`. Records with unknown incumbent status are not included in either group's denominator.

The winning rate is calculated as

$$\text{Win rate} = \frac{\text{Number of wins}}{\text{Total candidate records}} \times 100.$$

For incumbents, 57.88%; for non-incumbents, 40.17%. The observed difference is therefore 17.71 percentage points. The 95% Confidence intervals are 56.49–59.26% and 39.33–41.03%, while the 95% Confidence interval for the difference is 16.08–19.32 percentage points.
Incumbents had a substantially higher observed winning rate than non-incumbents in the analysed dataset. The confidence interval for the difference lies entirely above zero, supporting the presence of a clear difference in these observations.



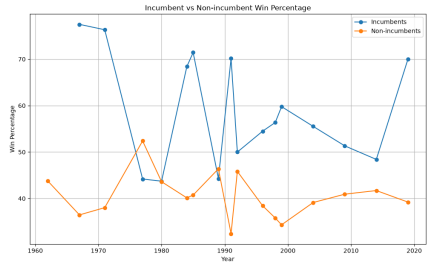
The bars show the two estimated win rates and their difference; the confidence intervals express uncertainty around the estimates.

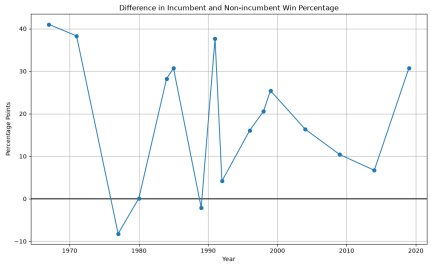
But an overall average can hide substantial variation. Does the incumbent advantage appear in every election year, or does it change over time?

What changes when the winning rates are examined election by election?

The year-wise line graph places election year on the horizontal axis and winning rate on the vertical axis, with separate lines for incumbents and non-incumbents. The difference graph shows $p_I - p_N$ in percentage points; values above zero favour incumbents, while values below zero favour non-incumbents.

The gap is not uniform. Non-incumbents have a higher rate in 1977 by 8.20 points and in 1989 by 2.09 points; the groups are almost equal in 1980. The incumbent advantage reaches 30.74 points in 2019. Thus, the aggregate advantage does not describe every election equally well, and annual sample sizes and missing classifications matter when interpreting the trend.

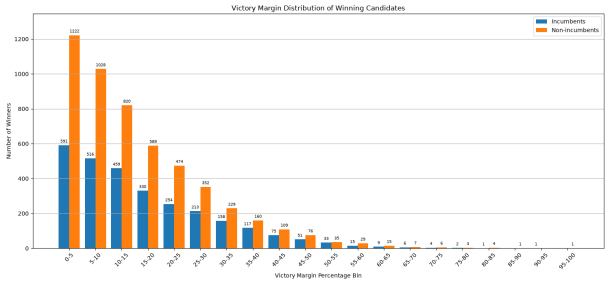


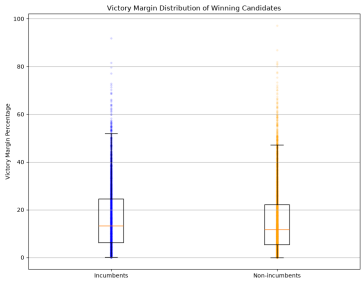


The year-wise pattern confirms that the aggregate gap varies. However, a win-rate comparison tells us only how often candidates win. It does not tell us how comfortably successful candidates win.

When incumbents win, do they also win by larger margins?

To examine the size of victories, the margin analysis considers winners only. Incumbent winners have a mean margin of 16.75% and a median of 13.19%, compared with 15.29% and 11.81% for non-incumbent winners. The mean difference is 1.47 percentage points. The standard deviations are 13.64 and 12.99 points, respectively.





These results suggest that incumbent candidates who win tend to secure slightly larger margins. However, this the graph tells us most of the candidates irrespective of incumbency fall under the 0-5% bins and this leads to analysis of closely contested elections as the mode is not convinving to tend towards either side of the main question . The margin comparison is read alongside the win-rate analysis.

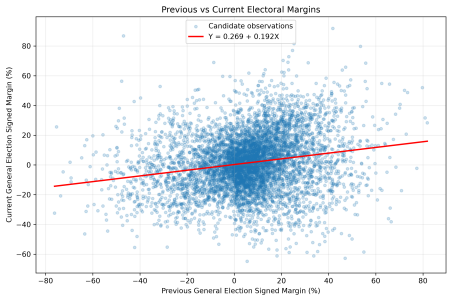
Even after studying how often candidates win and by how much, one question remains: is a candidate's previous electoral strength associated with their performance in a later contest?

Does a candidate's previous electoral performance carry forward?

A simple linear regression is used to examine the relationship between a candidate's previous signed margin and current signed margin. The predictor X is the previous signed margin, and the response Y is the current signed margin. Positive values indicate a win; negative values indicate a loss. The fitted model, based on 6,168 same-constituency matched observations, is

$$\hat{Y} = 0.2690 + 0.1916X, \qquad R^2 = 0.0389.$$

The slope means that a one-percentage-point increase in the previous margin is associated with an increase of about 0.192 percentage points in the predicted current margin. The scatterplot shows one point per matched candidate pair and the red line summarises the fitted relationship. . However, $R^2 = 0.0389$ means previous margin alone explains only about 3.89% of the variation in current margins. The relationship is positive but weak.



Even though the coefficients say that higher previous margin does tend to lead to a higher subsequent margin (the 0.1916 slope)—previous margin alone is only responsible for 3.89% of the variance in subsequent margins.

The remaining 96.11% of the variation is driven by other variables and this leads us to the next question.

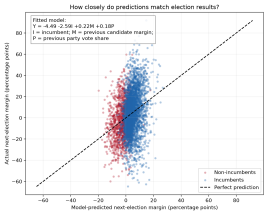
Previous performance matters, but it explains only a small part of later margin variation. Could party strength and incumbency status help explain what the simple model leaves out?

What changes when incumbency and party strength are considered together?

Multiple linear regression extends the simple model with three predictors: I (incumbency: 1 for incumbent and 0 otherwise), M (previous candidate signed margin), and P (the candidate's current party vote share in that constituency at its previous election). The response Y is the current signed margin. For 5,181 complete records, the fitted model is

$$\hat{Y} = -4.4527 - 2.4944I + 0.2237M + 0.1808P, \qquad R^2 = 0.0850.$$

The incumbency coefficient is -2.4944 (95% CI: -4.0522 to -0.9365; $p = 0.001705$): holding M and P constant, the model predicts a current signed margin about 2.49 points lower for incumbents. The coefficients of M and P are positive. With $R^2 = 0.0850$, the model explains 8.50% of the observed variation.



The graph compares predicted margins (horizontal axis) with actual margins (vertical axis); the dashed diagonal represents perfect prediction.

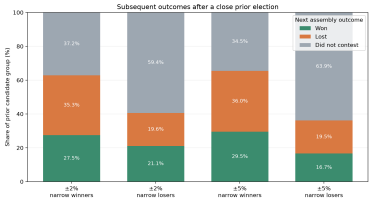
The negative adjusted incumbency coefficient differs from the positive raw win-rate gap. These findings are not contradictory: the win-rate comparison is an unadjusted group comparison, while the regression compares candidates with the same values of the included predictors. The coefficient is an adjusted association, not proof that incumbency reduces performance.
Could a closer comparison of candidates who barely won and barely lost help us examine the role of incumbency more carefully?

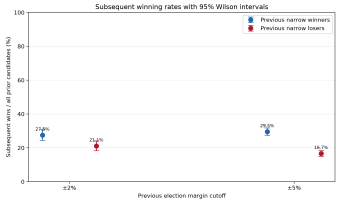
What happens after a closely contested previous election?

The close-election analysis follows candidates who narrowly won or lost a previous general election into the next Lok Sabha assembly.

For ±2%, previous narrow winners won (27.52%), against(21.08%) for narrow losers: a 6.44-point difference (95% CI: 1.98–10.86). Re-contestation rates were 62.84% versus 40.64%; conditional win rates among returners were 43.79% versus 51.88%.

For ±5%, all-candidate win rates were 29.52% versus 16.66% , a 12.86-point difference (95% CI: 10.10–15.61). Re-contestation rates were 65.52% versus 36.13%, while conditional win rates were 45.06% versus 46.11%.





Based on the descriptive analysis, incumbents clearly had a higher observed winning rate than non-incumbents.However, when we accounted for previous electoral margin and party vote share in the multiple regression, the incumbency coefficient became negative.

The close-election analysis added another dimension: previous narrow winners were more likely to win subsequently when all candidates were considered, but they were also much more likely to contest again. Among those who returned, the difference in winning rates was much smaller.

The final interpretation is that incumbency is associated with an overall electoral advantage, but this advantage is not uniform or straightforward. It varies across election years, depends on how electoral performance is measured, and is intertwined with previous candidate performance, party support and re-contesting behaviour.